MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 12000010

(BALLISTIC CONSTANT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Ballistic Constant Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assembly, Sight Current Generator, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

10545455	Receptacle
12000010	Circuit Card Assembly, Ballistic Constant
12000239	Sight Current Generator

PACKAGING DATA SHEET

12000010	Packaging of Circuit Card Assembly, Ballistic Constant

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

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3.1 Item definition.- This assembly is a single-ended printed circuit board consisting of two transistors and four operational amplifiers. The circuit card is used to sum, invert, and/or amplify the multiple inputs which make up the magnet and torque current voltages developed in the Sight Current Generator Drawing Number 12000239 in which the circuit card is assembled.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000010.

3.1.2 Government furnished property.- Not applicable.

3.2 Characteristics.

3.2.1 Performance.- The assembly shall be capable of performing the following requirements when assembled in accordance with Drawing Number 12000010.

a. Operational amplifier AR1 shall sum and invert the inputs required to develop the torque current voltage.

b. Operational amplifier AR2 shall amplify and invert the input required to drive the self test circuit.

c. Operational amplifier AR3 shall sum and invert the inputs required to develop the magnet current voltage.

d. Operational amplifier AR4 shall sum and invert four of the inputs required to develop the magnet current voltage.

e. Transistors Q1 and Q2 shall change the gain of two of the inputs to AR4.

3.2.1.1 Electrical requirements.- The offset voltage of each operational amplifier in the assembly shall be as specified in Table I with zero Vdc applied to the input pins. The input and output pins shall be as specified in Table I.

TABLE I - INPUT AND OUTPUT PINS

Amplifiers	Input Pins	Output Pins	Offset Voltage
AR1	2, 3, 10	5	± .020 Vdc
AR2	15	12	± .020 Vdc
AR4	8, 16, 20, 23, 26	19	± .020 Vdc

3.2.1.2 Input and output levels.

3.2.1.2.1 Input power.- Input power shall be supplied to the assembly as specified in Table II.

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TABLE II - INPUT POWER

PIN	VOLTAGE
4 and 25	+15 Vdc ± 2.0 percent
14	-15 Vdc ± 2.0 percent
1 and 28	Ground

3.2.1.2.2 Output levels.- The requirements specified in 3.2.1.2.2.1 through 3.2.1.2.2.4 shall be met when supplied with the voltage input (Vin) specified in Table II.

3.2.1.2.2.1 Pin 5 output.- The output between pin 5 and pin one shall be as specified in a, b, and c.

a. Vin (+0.1860) ± 1.0 percent with a Vin of -15 Vdc to +15 Vdc between pin 2 and pin one, when pins 3 and 10 are at zero Vdc.

b. Vin (-4.8020) ± 1.0 percent with a Vin of -2.7 Vdc to + 2.7 Vdc between pin 3 and pin one, when pins 2 and 10 are at zero Vdc.

c. Vin (-0.6317) ± 1.0 percent with a Vin of -13 Vdc to +13 Vdc between pin 10 and pin one, when pins 2 and 3 are at zero Vdc.

3.2.1.2.2.2 Pin 12 output.- The output between pin 12 and pin one shall be as specified in a and b.

a. As shown in Figure 1 when -1.0 Vdc is applied between pin 15 and pin one.

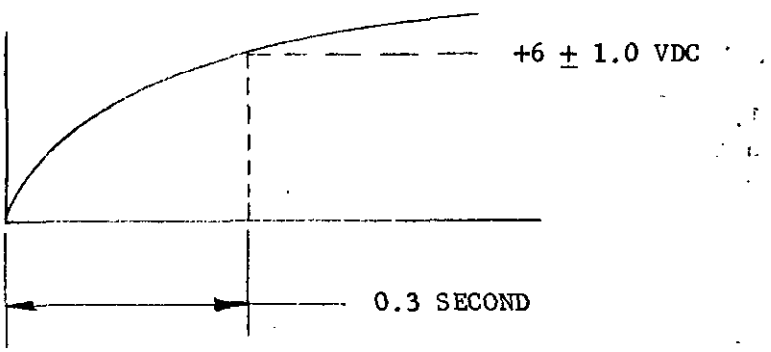


Figure 1.

b. Vin (-7.6278) ± 1.0 percent with a Vin of -1.7 Vdc to + 1.7 Vdc applied between pin 15 and pin one.

3.2.1.2.2.3 Pin 17 output.- The output between pin 17 and pin one shall be as specified in a, b, and c.

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a. Vin (-0.1281) + 0.150 Vdc ± 1.0 percent with a Vin of -13 Vdc to zero Vdc applied between pin 21 and pin one when pin 22 is at zero Vdc.

b. Vin (-1.9050) + 0.150 ± 1.0 percent with a Vin of 6.5 Vdc to zero Vdc applied between pin 22 and pin one when pin 21 is at zero Vdc.

3.2.1.2.2.4 Pin 19 output.- The output between pin 19 and pin one shall be as specified in a through g.

a. As shown in Figure 2 when -10 Vdc is applied between pin 20 and pin one.

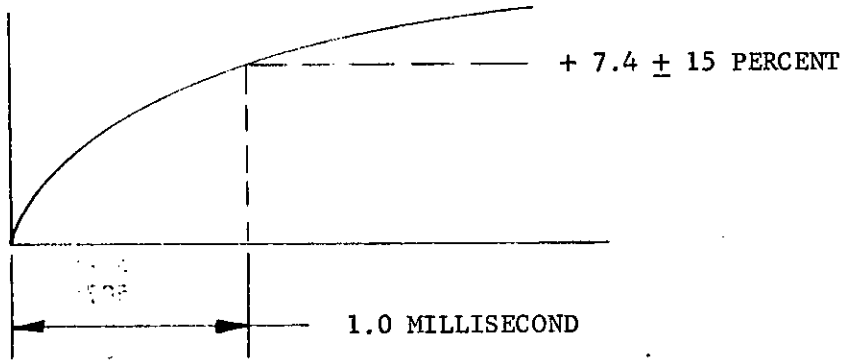


Figure 2.

b. Vin (-1.0000) ± 1.0 percent with a Vin of zero Vdc to +12 Vdc applied between pin 20 and pin one when pins 8, 16, 23 and 26 are at zero Vdc.

c. Vin (-0.0365) ± 5.0 percent with a Vin of -13 Vdc to zero Vdc applied between pin 8 and pin one when pins 16, 20, 23 and 26 are at zero Vdc.

d. Vin (-0.6608) ± 1.0 percent with a Vin of -13 Vdc to zero Vdc applied between pin 16 and pin one when pins 8, 20, and 26 are at zero Vdc and pin 23 is at -15 Vdc.

e. Vin (-0.2000) ± 1.0 percent with a Vin of -13 Vdc to zero Vdc applied between pin 16 and pin one when pins 8, 20, and 26 are at zero Vdc and pin 23 is at +15 Vdc.

f. Vin (-1.333) ± 1.0 percent with a Vin of zero Vdc to +8.9 Vdc applied between pin 26 and pin one when pins 8, 16, and 20 are at zero Vdc and pin 23 is at -15 Vdc.

g. Vin (-0.4274) ± 1.0 percent with a Vin of zero Vdc to +13 Vdc applied between pin 26 and pin one when pins 8, 16 and 20 are at zero Vdc and pin 23 is at +15 Vdc.

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3.2.1.3 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristics.- The physical characteristics shall be as specified in drawing 12000010 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 12000010 and associated lists.

3.3.2 Standards of manufacture.- The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-50785.

4. QUALITY ASSURANCE PROVISION

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment.

4.2.1.1 Power supply (PS1).- The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS2).- The power supply shall be -15 ± 0.1 Vdc, 100 ma or greater, ripple less than 10 mv rms or 100 mv peak-to-peak.

4.2.1.3 Power supply (PS3).- The power supply shall be zero ± 13 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 10 ma or greater.

4.2.1.4 Switches.- Three double pole single throw switches are required.

4.2.1.5 Oscilloscope.- The oscilloscope shall have voltages, time accuracy and resolution of 5.0 percent or better, with a rise time of 100 nanoseconds or better and with external triggering. Voltage sensitivity shall include the range from 0.05 volt/cm to 5.0 volts/cm.

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4.2.1.6 Digital voltmeter (DVM).- The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

4.2.2 Functional test.

4.2.2.1 Test connections.- Power supplies PSI and PS2 shall be connected to the assembly under test in accordance with Figure 3 using receptacle 10545455.

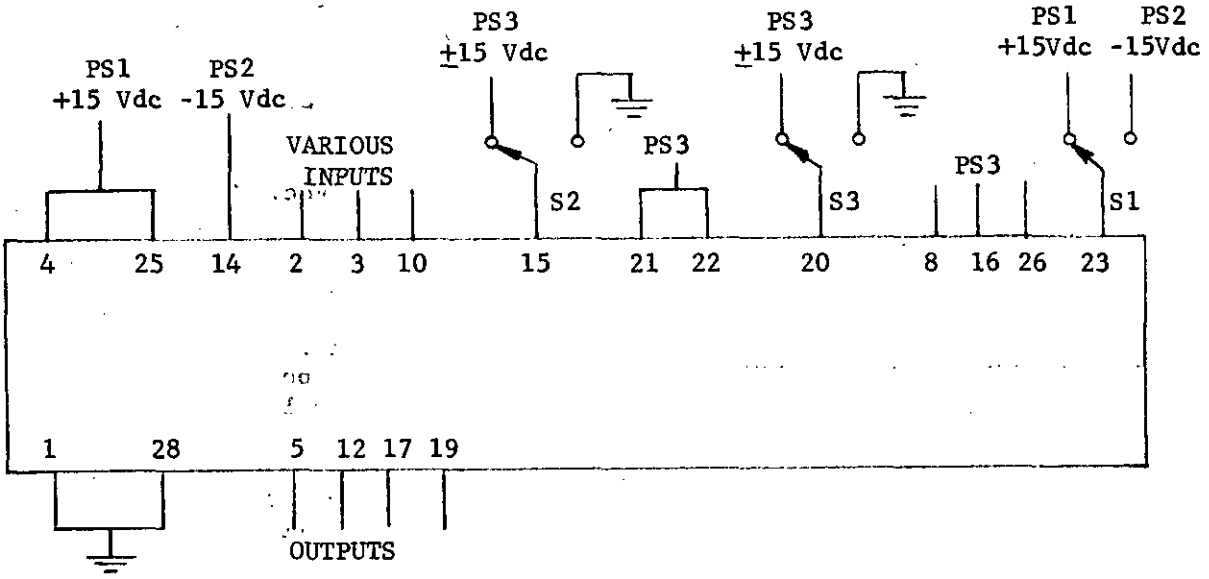


Figure 3. Test Connections

4.2.2.2 Test procedure.- Compliance with the requirements of 3.2.1.1 through 3.2.1.3 shall be verified as follows:

a. Connect and adjust PS3 as specified in Table III.

TABLE III - CONNECTIONS AND ADJUSTMENTS

Input Pin 2 Vdc	Input Pin 3 Vdc	Input Pin 10 Vdc	Output Pin 5 Vdc
+15	Ground	Ground	+2.790 ± 0.028
-15	Ground	Ground	-2.790 ± 0.028
Ground	Ground	Ground	0 ± 0.020
Ground	-2.7	Ground	12.965 ± 0.130
Ground	Ground	-13 Vdc	8.212 ± 0.082

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b. For steps 1, 2, and 3 connect PS3 to the -1.7 volt position of S2.

(1) Set S2 to the -1.7 volt position. The output measured at pin 12 with the DVM shall be 12.967 ± 0.130 Vdc.

(2) Set S2 to the ground position. The output measured at pin 12 with the DVM shall be 0 ± 0.020 Vdc.

(3) Adjust PS3 to -1 Vdc. Switch S2 from the ground position to the -1 Vdc position. The output measured at pin 12 with the oscilloscope shall be as shown in Figure 1.

c. Connect PS3 to pin 21. Adjust PS3 for -13 Vdc. The output measured at pin 17 shall be 1.815 ± 0.020 Vdc.

d. Connect PS3 to pin 22. Adjust PS3 for -6.5 Vdc. The output measured at pin 17 shall be 12.533 ± 0.125 Vdc.

e. Connect PS3 to pin 20. Adjust PS3 for +12.0 Vdc. The output measured at pin 19 shall be -12 ± 0.120 Vdc.

f. Connect PS3 to pin 8. Adjust PS3 for -13 Vdc. The output measured at pin 19 shall be 0.474 ± 0.025 Vdc.

g. Connect PS3 to pin 16. Set S1 to the -15 Vdc position. Adjust PS3 for -13 Vdc. The output measured at pin 19 shall be 8.581 ± 0.086 Vdc.

h. Connect PS3 to pin 16. Set S1 to the +15 Vdc position. Adjust PS3 for -13 Vdc. The output measured at pin 19 shall be 2.6 ± 0.026 Vdc.

i. Connect PS3 to pin 26. Set S1 to the -15 Vdc position. Adjust PS3 for +8.9 Vdc. The output measured at pin 19 shall be -11.864 ± 0.120 Vdc.

j. Connect PS3 to pin 26. Set S1 to the +15 Vdc position. Adjust PS3 for +13 Vdc. The voltage measured at pin 19 with the DVM shall be -5.556 ± 0.056 Vdc.

k. Ground pins 8, 16, and 23. Set S3 to the ground position. The output measured at pin 19 shall be 0 ± 0.020 Vdc.

l. Adjust PS3 to -10 Vdc. Switch S3 from the ground position to the -10 Vdc position. The output measured at pin 19 with the oscilloscope shall be as shown in Figure 2.

5. PREPARATION FOR DELIVERY

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5.1 Packaging.- Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 12000010.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

Custodian:

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Preparing Activity:

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Project No. 1285-A433

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DOCUMENT IDENTIFIER AND TITLE MIL-C-50785/14(MU) Circuit Card Assembly: (Ballistic Constant)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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